GOVERNMENT ENGINEERING COLLEGE, RAJKOT

GENERAL DEPARTMENT

Subject: **DMGT (BE04000261)** Semester: **4^{th}** Branch: **CE/AIDS**
Tutorial-**02 (CO-2)** Term Date: 01/01/2026 to 06/05/2026

Ex-1 Explain the roster notation and set builder notation of sets with examples.

Ex-2 What is the power set? State the relation between the cardinalities of a finite set and its power set.

Ex-3 Define the Cartesian product of two sets and give an example.

Ex-4 It is known that in university 60% of professors play tennis, 50% of them play bridge,70% jog, 20% play tennis and bridge, 40% play bridge and jog, 30% play tennis and jog. If someone claimed that 20% professors jog and play tennis and play bridge, would you believe his claim? Why?

Ex-5 Among integers 1 to 1000,

(a) How many of them are not divisible by 3 nor by 5 nor by 7?
(b) How many are not divisible by 5 or 7 but divisible by 3?

Ex-6 A survey among 1000 people, 595 are democrats, 595 wear glasses and 550 like ice cream. 395 of them are democrats who wear glasses 350 of them are democrats who like ice cream. 400 out of them wear glasses and like ice cream and 250 all the three.

(a) How many of them are not democrats, do not wear glasses and do not like ice-cream?
(b) How many of them are democrats who do not wear glasses and do not like ice-cream?

Ex-7 Find a formula for the general term a_n of the Fibonacci sequence 0,1,1,2,3,5,8,13,...with the initial conditions $a_0 = 0$ $a_1 = 1$

Ex-8 If $A = \{a, b\}$, $B = \{c, d\}$ and $C = \{e, f\}$ then find (i) $(A \times B) \cup (B \times C)$, (ii) $A \times (B \cup C)$

Ex-9 If $A = \{x \mid x = 3^n,\ n \le 6, n \in N\}$ $B = \{x \mid x = 9^n,\ n \le 4, n \in N\}$ then find $A \bigcup B$, $A \bigcap B$ and $A - B$.

Ex-10 In a survey of 400 students in a school, 100 were found as drinking apple juice, 150 as drinking orange juice and 75 drinking both apple and orange juice. Find how many students drink neither apple nor orange juice.

Ex-11 Find n if ${}^{n}C_3 + {}^{n+2}C_3 = {}^{n}P_3$

Ex-12 List the domains of the following sets: $N = \{1,2,3,....\}$

(a) $A = \{x : x \in N, 2 < x < 14\}$
(b) $B = \{x : x \in N, x \text{ is odd}, x < 15\}$
(c) $C = \{x : x \in N, 5 + x = 3\}$

Ex-13 How many permutations of the letters A B C D E F G contain (a) the string BCD, (b) the string CFGA, (c) the strings BA and GF, (d) the strings ABC and DE, (e) the strings ABC and CDE, (f) the strings CBA and BED?